Ex.No:1	Study of various Network Commands used in Linux and Windows
23.07.26	

AIM:

To study of various Network commands used in Linux and Windows.

Commands:

1) arp -a:

ARP is short form of address resolution protocol, It will show the IP address of your computer along with IP address and MAC address of your router.

Output:

Internet Address	Physical address
172.16.8.149	172:16:8:13:f3:fe
172.16.9.212	7C:57:58:38:da:a3
172.16.8.186	7C:57:58:38:e6:24
172.16.8.249	00:27:0e:13:f5:77
172.16.11.207	7C:57:58:38:46:87.

2, hostname:

This is the simplest of all TCP/IP commands. It simplify displays the name of your computer.

Output:

localhost.localdomain

3. ipconfig /all:

This command displays detailed configuration information about your TCP/IP connection including Router, Gateway, DNS, DHCP and type of ethernet adopter in your system.

Output:

DHCP Server : 192.168.112.254

DHCPv6 IAID: 587223126

DHCPv6 Client DUID: 00-01-00-01-BD-D7-41-B3-7C-57-58

Primary WINS Server: 192.168.112.2

NetBIOS Over Tcpip : Enabled.

4) netstat: (network statistics)

netstat displays a variety of statistics about a computers active TCP/IP connections. It is a command line tool for monitoring network connections both incoming and outgoing as well as viewing routing tables, interface statistics, etc.

netstat -r

Output:

Interface List

2. 7c 57 58 35 10 fb

17. 00 50 56 c0 00 01

11. 00 50 56 c0 00 08

1 Software Loopback Interface 1.

5. nslookup: (name server lookup) is a tool used to perform DNS lookups in Linux. It is used to display DNS details, such as the IP address of a particular computer, the MX records for a domain or the NS servers of a domain nslookup can operates in two modes: interactive and non-interactive.

Eg: nslookup www.google.com.

Output:

Server unknown

Address: 172.16.8.1

Non-authorative answer

Name: Star mini or ~~facebook~~ google.com.

Address: 2a03 2880 f137 182; 6a3e:

booc:0:25de 157.240.192.35

Aliases ww.google.com.

Linux Commands:

1) arp -a:

gateway (172.16.8.1) at 7c 5a 1c cf be 45 [ether] on enp2s0

2) ifconfig:

enp2s0: flags = 4163 <UP, BROADCAST, RUNNING, MULTICAST > mtu 1500

inet 172.16.8.112 netmask 255 255 252 broadcast 172.16.11.255

ether 50:9a:4c:35:0f:45

txqueuelen 1000 (Ethernet)

wlp3s0: flags = 4099 <UP, BROADCAST, MULTICAST > mtu 1500

ether 4a:25:d4:c1:93:ab

tx queuelen 1000 (Ethernet)

3. hostname:
   localhost localdomain

4. nmblookup -A 172.16.8.1
   Looking up status of 172.16.8.1
   No reply from 172.16.8.1

5. netstat -r
   Kernel IP routing table
   Destination Gateway Genmask
   Flags MSS in Window
   default gateway 0000 UG 0000
   172.16.8.0 0.0.0.0 255.255.252.0 U
   00 00

Important Linux Networking Commands

1) ip:

Every administrator will need in daily work from setting up new system

Eg: ip address show

link (loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00)

inet 127.0.0.1 18 scope host lo.

2. mtr:

Serves as a network diagnostic tool.

mtr google.com

Packets

loss % snt Last avg Boot wts

0.0% 17.8 0.1 0.1 0.0 0%.

3. tcpdump: designed for capturing and displaying packets

output: tcpdump -D

4. ping: Tool to verify IP-level connectivity to another TCP/IP computer by sending Internet control message protocol Echo request messages.

output: ping localhost

ping localhost (localhost (::1) 56 data bytes 64 bytes from localhost (::1) time = 0.053 ms.

Observation:

1, Which command is used to find the reachability of a host machine from your device?

ping command <hostname/IP>

ping is primary TCP command used to troubleshoot connectivity reachability.

2. Which command will be give the details of hops taken by a packet to reach its destination?

mtr (Matt's traceroute)

It show route from a computer to a specified host mtr provides a lot of statistics about each hop such as response time and percentage.

3, Which command displays the ip configuration of your machine?

ip <options> <object> <command>

Ip command can show address information manipulate routing plus display network various devices network various devices, interfaces & funnels.

4) Which command displays the TCP port status in your machine?

netstat

It displays variety of statistic about a computer active TCP/IP connection.

5, Write the modify the ip configuration in a linux machine?

* Assigning IP address to interfaces

ip address add 192.168.1.254/24 dev enp503

* Deleting IP address

ip address del 192.168.1.254/24 dev enp503

Result:

Thus, the Network commands using Linux and Windows are successfully executed and the output is verified.